ENGINEERING DESIGN DEPARTMENT · OPERATING MODEL PROPOSAL

From Reactive Engineering to Predictable Engineering

Protecting engineering capacity through controlled demand management.

Today: random demand

Single entry point

Committed work packages

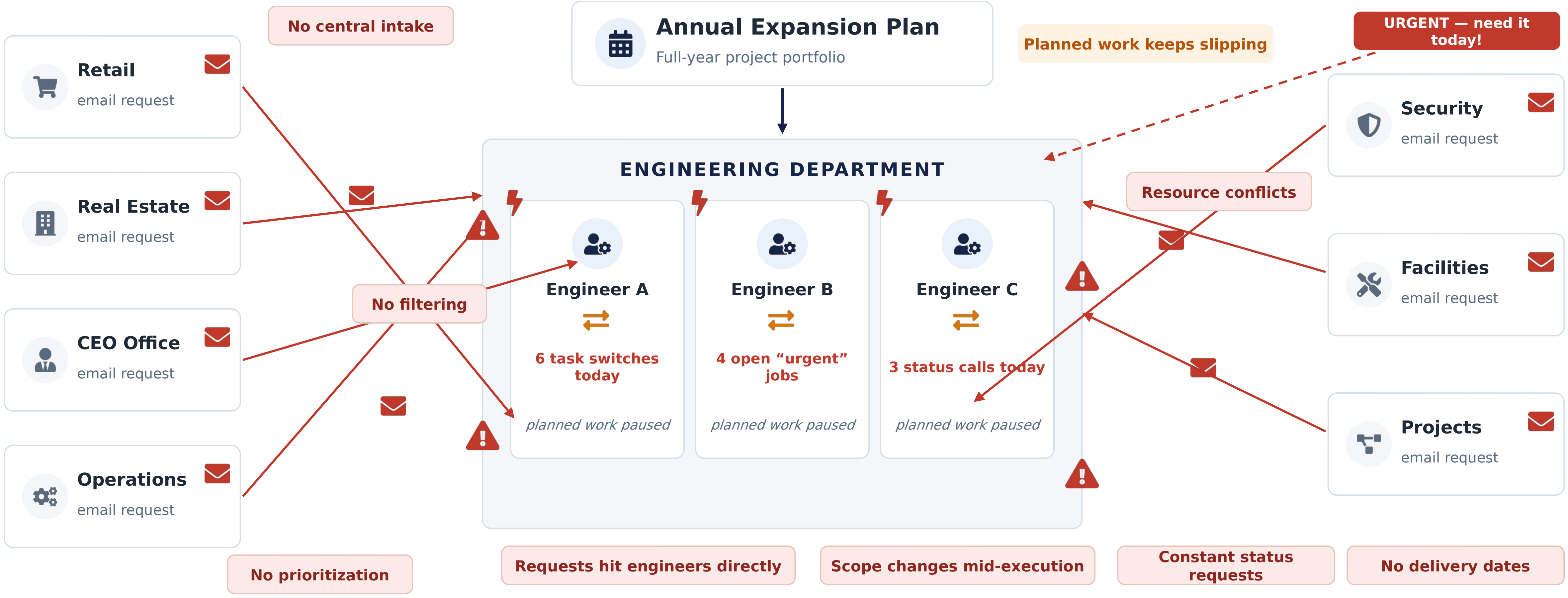
Predictable delivery

Executive Briefing · August 2026

AS-IS · CURRENT STATE

Simulation 1 — how work happens today

Work reaches engineers from every direction



PAIN POINTS

- ✗ Constant interruptions
- ✗ No visibility
- ✗ No workload control
- ✗ No commitment dates
- ✗ Low productivity
- ✗ Context switching
- ✗ Coordinators, not designers

TO-BE · FUTURE STATE

Simulation 2 — the proposed operating model

One front door — the Engineering Design Delivery System

ALL REQUESTS

- Retail
- Real Estate
- CEO Office
- Operations
- Security
- Facilities
- Projects

Requests keep arriving — 24/7

Annual Engineering Plan

ENGINEERING INTAKE

Single entry point — no engineer receives requests directly

ENGINEERING PIPELINE

REQ P1	REQ P2	REQ P3	REQ P4	REQ P5

Every request is logged, scored and visible — nothing gets lost

WEEKLY ENGINEERING PLANNING SESSION

Management + stakeholders decide: Priorities · Scope · Capacity · Delivery commitments

RELEASED: Work Package → Engineering Team

EACH REQUEST CARRIES

- Priority
- Business Value
- Required Date
- Complexity
- Estimated Effort
- Owner
- Status

Only after planning is work released — nothing bypasses the gate.

TO-BE · FUTURE STATE

From first email to delivered design

The Engineering Pipeline — every request follows one path

DEMAND INTAKE · 1–3 | **GOVERNANCE & PLANNING · 4–7** | **EXECUTION & DELIVERY · 8–10**

01 **Request Received**
Logged at central intake

02 **Intake Review**
Is it engineering work?

03 **Information Validation**
Complete brief? Data ready?

04 **Engineering Pipeline**
Queued & visible to all

05 **Prioritization**
Priority × business value

06 **Capacity Check**
Free slots per capability?

07 **Work Package Creation**
Locked scope for the cycle

08 **Engineering Execution**
Focused design work

09 **QA**
Design review & checks

10 **Delivery**
Committed date met

Stages 1–7 absorb and govern demand. Engineers only ever see stages 8–10 — released, planned, capacity-checked work.

This is an engineering delivery pipeline — inspired by flow principles, designed for engineering design work (not a software process).

TO-BE · FUTURE STATE

Released by capacity — not by urgency

Balancing unlimited demand against fixed capacity

INCOMING DEMAND

Continuous · unbounded · from every department

27 new engineering requests received this week

REQ-112 · Retail	REQ-113 · Security	REQ-114 · CEO Office
REQ-115 · Facilities	REQ-116 · Operations	REQ-117 · Projects
REQ-118 · Real Estate	REQ-119 · Retail	REQ-120 · Operations
REQ-121 · Facilities	REQ-122 · Security	REQ-123 · Retail

...and more arriving every day

WEEKLY PLANNING SESSION

RELEASE DECISION

Priority × Business Value × Available Capacity

✗ not urgency ✗ not who shouts loudest

Demand exceeds capacity every week — planning decides what wins, openly.

ENGINEERING CAPACITY

Fixed weekly slots per engineering capability

Cost Estimation	3 / wk
Architectural Design	2 / wk
Site Evaluation	2 / wk
Space Planning	2 / wk
BOQ Preparation	3 / wk
Design Review	4 / wk

Total capacity: **16 project slots / week**

Example: Cost Estimation can complete 3 projects this week → only 3 are released. Everything else waits safely in the pipeline.

TO-BE · FUTURE STATE

Locked scope · protected team · committed date

A released Work Package is a commitment

WEEK 1 PLANNING

Pipeline reviewed
Capacity checked
Package approved

→

WORK PACKAGE A

- ✓ **Project 1 — Architectural Design**
- ✓ **Project 2 — Cost Estimation**
- ✓ **Project 3 — Site Evaluation**

Scope locked at release — no additions mid-cycle

→

ENGINEERING TEAM

Works only on
Package A during
the agreed cycle

→

COMMITTED DELIVERY

End of Week 4 — a date
stakeholders can trust

New urgent request arrives mid-cycle → **EXCEPTION GOVERNANCE**

approved → enters the package

declined → waits in the pipeline

THE PACKAGE MOVES AS ONE STABLE UNIT

Package A — stable scope from release to delivery →

WEEK 1 | WEEK 2 | WEEK 3 | WEEK 4 | DELIVERY

No new work enters a released package without governance approval — that is how delivery dates become predictable.

TO-BE · FUTURE STATE

Swimlane — one full cycle

The engineering work cycle — focus protected for four weeks

INCOMING REQUESTS — CONTINUOUS

ENGINEERING PIPELINE — requests wait here for the next planning session

INTERRUPTION BARRIER — nothing crosses mid-cycle

ENGINEERING TEAM — PROTECTED

WEEK 1 — **PLANNING**

WEEK 2 — **ENGINEERING DESIGN**

WEEK 3 — **ENGINEERING DESIGN**

WEEK 4 — **QA**

DELIVERY PACKAGE

✓ **Engineers stay focused — zero interruptions inside the cycle**

next cycle begins — the pipeline feeds Week 1 planning

Demand never stops — but it never touches an engineer mid-cycle.

AS-IS vs TO-BE

Built to be understood in 30 seconds

The difference at a glance

TODAY — REACTIVE

- Random arrows everywhere
- Constant interruptions
- Emails from all directions
- Broken workflow
- Direction changes daily
- No commitment dates

TOMORROW — PREDICTABLE

One straight line from request to delivery

- One single flow
- Controlled queue
- Stable work packages
- Central governance
- Clear priorities
- Predictable delivery

The same demand — run through a different system.

MANAGEMENT BENEFITS

12 dimensions, one decision

What management gains

DIMENSION	CURRENT STATE — REACTIVE	FUTURE STATE — PREDICTABLE
Request Intake	✗ Email to any engineer, any time	✓ **Single central intake — one front door**
Prioritization	✗ First come / loudest voice wins	✓ **Scored by priority, value and capacity**
Visibility	✗ Work hides in personal inboxes	✓ **Full pipeline visible to management**
Capacity Planning	✗ Unknown — always overloaded	✓ **Fixed weekly slots per capability**
Engineer Focus	✗ Interrupted daily, constant switching	✓ **Protected inside the work package**
Delivery Predictability	✗ Dates slip continuously	✓ **Committed date per work package**
Governance	✗ Ad-hoc decisions, no gate	✓ **Weekly planning + exception process**
Customer Expectations	✗ Everything “urgent”, nothing certain	✓ **Clear commitments and queue position**
Resource Utilization	✗ Fragmented by context switching	✓ **Full slots spent on planned work**
Quality	✗ Rushed under pressure	✓ **Dedicated QA week in every cycle**
Management Reporting	✗ Manual chasing for status	✓ **Pipeline and package status on demand**
Workload Balancing	✗ Random — whoever gets asked	✓ **Matched to capability capacity**

Same engineers. Same demand. A different operating system for work.

THE TRANSFORMATION

From Reactive Engineering to Predictable Engineering.

Protecting engineering capacity through controlled demand management.

01

SINGLE INTAKE

Every request enters through one front door. Engineers are never interrupted directly — the system absorbs the demand.

02

CAPACITY-BASED RELEASE

Work is released weekly against real capability capacity — priority and business value decide, not urgency.

03

COMMITTED WORK PACKAGES

Locked scope per cycle turns engineering effort into delivery dates stakeholders can trust.

The decision requested: approve the Engineering Design Delivery System — stand up Central Intake and the Weekly Planning Session.